Lecture 1 (September 2nd)

Remark. The main subject of this course is Galois theory. The motivation of this theory is to find a general solution of polynomial equations. For example, the general solution of the quadratic equation

$$ax^2 + bx + c \;=\; 0$$

where $a \neq 0$, b and c are real numbers is given by

$$x \;=\; \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

To be more precise, we want to express a general solution of the equation

$$a_n x^n + a_{n-1} x^{n-1} + \ldots + a_0 = 0$$

where $a_i \in \boldsymbol{Q}$ and $a_n \neq 0$ using 4 basic operations (namely, $\pm$, $\times$, and $\div$), and n-th roots. Today, we know that, through Galois theory, this is not possible for $n \geq 5$.

In essence, Galois theory connects the theory of fields and the theory of groups. Consider the following example. The polynomial

$$x^4 + x^3 + x^2 + x + 1 = 0$$

We know how to solve this,

$$(x-1)(x^4 + x^3 + x^2 + x + 1) = x^5 - 1 = 0$$

and thus we choose a primitive 5th root of unity (say $\xi = \exp(2\pi i/5)$). Then the solutions are

$$(\xi^1, \xi^2, \xi^3, \xi^4)$$

We see that this group is homomorphic to $\boldsymbol{Z}/5\boldsymbol{Z}$, and that there's a certain relationship between the roots of this polynomial and a certain group.

Chapter 13. Field Extensions

Definition. (13.1) Let K be a field. A field extension of K is another field F endowed with an injective ring homomorphism $\phi : K \to F$. You are essentially identifying the field with a subfield of a larger ring F.

Remark. From now on, we use the following conventions:

(i) Every ring is a commutative ring with 1

(ii) A ring homomorphism sends 1 to 1

Remark. There is at most one ring homomorphism between two fields. Notice that $\ker \phi$ is an ideal (additive subgroup with the absorption property) of K, so that $\ker \phi$ is either 0 or K. Therefore, ϕ must be injective.

Lecture 2 (September 4th)

Definition. (13.1) Let K be a field. A field extension of K is a pair $(F, i : K \to F)$ where F is a field and i is an injective (ring) homomorphism. We draw field extension tower diagrams like the following:

$$\begin{array}{c} F \\ | \\ K \end{array}$$

Example. (13.2 $*$) (Algebraic extension) Let $f(t) \in \boldsymbol{Q}[t]$ be a nonzero irreducible polynomial. Then $\boldsymbol{Q}[t] \,/\, (f(t))$ is a field, and the composite map

$$\boldsymbol{Q} \to \boldsymbol{Q}[t] \to \boldsymbol{Q}[t] \,/\, (f(t))$$

is a field extension. This is the general way we would be obtaining field extensions.

$$\begin{array}{c} \boldsymbol{Q}[t] \,/\, (f(t)) \\ | \\ \boldsymbol{Q} \end{array}$$

Proof. The set $\boldsymbol{Q}[t] \,/\, (f(t))$ is a maximal ideal and since $f(t)$ is a nonzero irreducible element of $\boldsymbol{Q}[t]$ which is principle ideal domain. Therefore, the quotient $\boldsymbol{Q}[t] \,/\, (f(t))$ is a field. $\square$

Example. (13.3) (Transcendental extension) Let $\boldsymbol{Q}(t)$ denote the field of fractions of $\boldsymbol{Q}[t]$. Then the composition map $\boldsymbol{Q} \to \boldsymbol{Q}[t] \to \boldsymbol{Q}(t)$ determines a field extension.

Definition. (13.6) The characteristic of a field K ($\operatorname{char}(K)$) is the smallest nonnegative generator of the kernel of the unique ring homomorphism $\phi : \boldsymbol{Z} \to K$.

Lemma. (13.7) Let $F \,/\, K$ be a field extension. Then

$$\operatorname{char}(K) = \operatorname{char}(F)$$

Lemma. (13.8) Let F be a field. Then F contains a unqiue copy of $\boldsymbol{Q}$ or $\boldsymbol{Z} \,/\, p\boldsymbol{Z}$, where p is prime.

13.2 Finite Extensions and the Degree of an Extension

Definition. (13.10) The degree of a field extension F/K, denoted $[F : K]$, is the dimension of F as a vector space over K.

$$[F : K] = \dim_K(F)$$

Remark. If $i : K \to F$ is a field extension, then we can view F as a K-vector space through the map i.

Definition. (13.11) A field extension is finite if $[F : K] < \infty$.

Definition. (algebraic and transcendental elements) Let F/K be a field extension, and let $\alpha \in F$. We will say that α is algebraic over K if there exists a nonzero polynomial $f(t) \in K[t]$ such that $f(\alpha) = 0$. The element $\alpha \in F$ is transcendental if not algebraic.

Example. Consider

$$\begin{array}{c} \boldsymbol{C} \\ \Big| \\ \boldsymbol{R} \end{array}$$

(i) Is $i \in \boldsymbol{C}$ algebraic over $\boldsymbol{R}$? Yes! i is a zero of $t^2 + 1 \in \boldsymbol{R}[t]$.

(ii) $\sqrt{2} \in \boldsymbol{R}$ is algebraic over $\boldsymbol{Q}$

(iii) Also, $\pi, e \in \boldsymbol{C}$ is transcendental over $\boldsymbol{Q}$.

(iv) Every element α of K is algebraic over K as we have the polynomial

$$t - \alpha \in K[t]$$

Definition. (13.20) Let F/K be a field extension. If $\alpha \in F$ is algebraic, the minimal polynomial of α over K is the unique monic irreducible polynomial $p(t) \in K[t]$ such that $f(\alpha) = 0$.

Proof. We show that the minimal polynomial exists and is unique. We will achieve this by showing that it is exactly the generator of the kernel of the evaluation map. First consider the evaluation homomorphism

$$\mathrm{ev}_\alpha : K[t] \to F$$

where

$$f(t) \mapsto f(\alpha) \qquad \& \qquad \sum_{i=0}^{n} a_i t^i \mapsto \sum_{i=0}^{n} a_i \alpha^i$$

Observe that it is indeed a homomorphism as it preserves addition, multiplication, and the identity element. Then, notice the following:

(I) The kernal of the map is automatically an ideal, and is of the form $(p(t))$ as $K[t]$ is a principle ideal domain.

(II) By the 1st isomorphism theorem, the following isomorhpism holds

$$K[t]\,/\,\ker(\mathrm{ev}_\alpha) = K[t]\,/\,(p(t)) \;\cong\; \mathrm{im}(\mathrm{ev}_\alpha) = K[\alpha] \subset F$$

This tells us that $K[\alpha]$ is a subring of F, $K[\alpha]$ is therefore a subring and an integral domain, and ultimately that $(p(t))$ is a prime ideal and that $p(t)$ is an irreducible element (we know that it is nonzero because α is algebraic).

(III) As generators are unique up to multiplication by a unit, we might as well suppose that the polynomial $p(t)$ is monic.

(IV) We lastly need to prove uniqueness. Suppose that there's another irreducible monic polynomial such that $q(t) = 0$. As $q(t) \in \ker(\mathrm{ev}_\alpha) = (p(t))$, $q(t) = p(t) \cdot r(t)$ and as $q(t)$ is irreducible, $r(t)$ has to be a unit. Lastly, $r(t)$ has to be 1 as both $q(t)$ and $p(t)$ are monic polynomials.

$$r(t) \in (K[t])^\times = K^\times = K \qquad \& \qquad p(t) = q(t)$$

$\square$

Lecture 3 (September 9th)

Definition. We let $K[\alpha]$ denote the subring generated by K and α and $K(\alpha)$ denote the subfield generated by K and α as seen below.

$$K[\alpha] = \{f(\alpha) \in F \mid f(t) \in K[t]\}$$

and

$$K(\alpha) = \left\{\frac{f(\alpha)}{g(\alpha)} \in F \mid f(t), g(t) \in K[t] \; g(\alpha) \neq 0\right\}$$

Recall. Take a principle ideal domain (PID) R and a nonzero $p \in R$. The following are equivalent

(i) p is prime

(ii) $(p) \subset R$ is prime

(iii) $(p) \subset R$ is maximal

(iv) p is irreducible

LECTURE 5 (SEPTEMBER 16TH)

Recall. Last time, we have dealt with $K[x]/(p(t))$ and minimal polynomials. Today, we will learn about properties of finite and algebraic extensions.

Definition. (Simple field extensions) A field extension F/K is simple if $F = K(\alpha)$ for some $\alpha \in F$, where $K(\alpha)$ denotes the smallest subfield of F containing $K \cup \{\alpha\}$. Notice that the field extension is constructed from simply adding a single element.

Remark. (Notation) $K(\alpha)$ is the smallest subfield, and $K[\alpha]$ is the smallest subring of F containing $K \cup \{\alpha\}$. Notice that these are equivalent to the definitions introduced last class due to inclusiveness and minimality.

Remark.

(i) $\alpha \in F$ is algebraic over K if and only if

$$\ker\Big(\mathrm{ev}_\alpha : K[t] \to F : f(t) \mapsto f(\alpha)\Big)$$

is nontrivial.

(ii) If α is algebraic over K, then $K[\alpha] = K(\alpha)$. In particular, every element of $K(\alpha)$ can be expressed as a polynomial in α with coefficient in K.

Proof. Consider $\alpha \in F$.

$$\ker(\mathrm{ev}_\alpha) = (p(t))$$

and due to the first isomorphic theorem,

$$K[t]/(p(t)) \cong K[\alpha]$$

If α is algebraic over K, then $(p(t))$ must be a maximal ideal of $K[t]$. This implies that $K[\alpha] = K(\alpha)$. ☐

Proposition. Let $K \subset K(\alpha) = F$ be a simple extension. Then either

(i) $F \cong K[t]/(p(t))$ for an irreducible monic polynomial $p(t) \in K[t]$ such that $p(\alpha) = 0$

(ii) $F \cong K(t)$, the field of fractions of $K[t]$

In the first case, $[F : K] = \deg(p(t))$. In the second case, $[F : K] = \infty$.

Example. (13.21) One example of a simple extension is

$$\alpha = 2^{1/5} \in \boldsymbol{Q}(2^{1/5})$$

Notice that π is algebraic in $\boldsymbol{R}$ but is not algebraic in $\boldsymbol{Q}$.

Theorem. (7.11) Let K be a field and $f(x)$ be a nonzero polynomial of degree n. Then,

$$K[x]/(f(x))$$

is a n-dimensional vector space over K.

Proof. Notice that the following set is a basis for $K[x]/(f(x))$

$$\{1, x, \dots, x^{n-1}\}$$

$\square$

Remark. (Notation) The minimal polynomial of α over K is denoted $\mathrm{irr}(\alpha, K)$.

13.4 Algebraic Extensions

Definition. (13.30) A field extension F/K is algebraic if every element of F is algebraic over K.

Proposition. (13.31) Every finite extension F/K is algebraic.

Proof. Fix an element $\alpha \in F$. Set $\dim_K F = n$. Then

$$\{1, \alpha, \alpha^2, \dots, \alpha^n\}$$

is linearly dependent over K, that is, $a_0 + a_1\alpha + \dots + a_n\alpha^n = 0$ for some $(a_0, \dots, a_n) \neq \mathbf{0} \in K^{n+1}$. Take

$$f(t) = a_0 + a_1 t + \dots + a_n t^n$$

in $K[t]$ and it is a polynomial with coefficients in K such that $f(\alpha) = 0$. $\square$

Proposition. (13.13) Let $K \subset E \subset F$ be field extensions. If E/K and F/E are both finite, then F/K is also finite. Furthermore,

$$[F : K] = [F : E][E : K]$$

Proof. Let $\{f_i\}_{i=1}^n$ be an E-basis of F and $\{e_j\}_{j=1}^n$ be a K-basis of E. Then $\{f_i e_j\}$ is a K-basis of F. The rest is for you! $\square$

Remark. If $\alpha \in F$ is algebraic over K, then $K(\alpha)/K$ is finite, hence algebraic. This is because

$$\dim_K K(\alpha) = \deg\Big(\operatorname{irr}(\alpha, K)\Big)$$

Here,

$$\dim_K K(\alpha) = \dim_K K[t]/(\operatorname{irr}(\alpha, K))$$

as

$$K[t]/(\operatorname{irr}(\alpha, K)) \cong K[\alpha] = K(\alpha)$$

due to the 3rd isomorphism theorem.

Proposition. A field extension $F \,/\, K$ is finite if and only if there are algebraic elements $\alpha_1, \alpha_2, \ldots, \alpha_r$ over K such that $F = K(\alpha_1, \alpha_2, \ldots, \alpha_r)$.

$$\begin{tikzcd}[arrows=dash]
F \arrow[dd] \arrow[dr] & \\
& K(\alpha_1, \alpha_2, \ldots, \alpha_n) \arrow[dl] \\
K &
\end{tikzcd}$$

Proof. (if) Suppose we are given algebraic elements $\alpha_1, \ldots, \alpha_r$ over K such that $F = K(\alpha_1, \ldots, \alpha_r)$. Consider the tower of finite towers

$$\begin{tikzcd}[arrows=dash]
F = K(\alpha_1, \ldots, \alpha_r) \arrow[d] \\
(K(\alpha_1, \alpha_2))(\alpha_3) \arrow[d] \\
(K(\alpha_1))(\alpha_2) = K(\alpha_1, \alpha_2) \arrow[d] \\
K(\alpha_1) \arrow[d] \\
K
\end{tikzcd}$$

$\square$

Lecture 6 (September 18th)

Remark. Las time, we learned about a tower of finite extensions. Today, we will learn about towers of algebraic extensions and constructability.

Proposition. If $\alpha \in F$ is algebraic over K, then

$$[K(\alpha) : K] = \deg(\mathrm{irr}(\alpha, K))$$

Proposition. (13.13) If $K \subset E$ and $E \subset F$ are finite extensions, then so is $K \subset F$. Furthermore,

$$[F : K] = [F : E][E : K]$$

Remark. The converse is also true.

Proposition. An extension $F \,/\, K$ is finite if and only if there are algebraic elements $\alpha_1, \ldots, \alpha_r$ over K such that $F = K(\alpha_1, \ldots, \alpha_r)$.

Proof. (only if) Assume that $F \,/\, K$ is finite. Then $F = K(\alpha_1, \ldots, \alpha_r)$ where $\{\alpha_1, \ldots, \alpha_k\}$ is a K-basis of F.

(if) If $r = 1$, then $K(\alpha_1) \,/\, K$ is finite by the proposition above. Inductively, we may assume that $r = 2$. In this case, consider the tower

$$\begin{array}{c} K(\alpha_1, \alpha_2) \\ | \\ K(\alpha_1) \\ | \\ K \end{array}$$

Since α_2 is algebraic over $K(\alpha_1)$, it follows from the above proposition that $K(\alpha_1, \alpha_2) = \Big(K(\alpha_1)\Big)(\alpha_2)$ is finite over $K(\alpha_1)$. Using proposition (13.13), we see that $K(\alpha_1, \alpha_2) \,/\, K$ is finite ($K(\alpha_1) \,/\, K$ is finite by the proposition above). $\square$

Proposition. (13.35) If $K \subset E$ and $E \subset F$ are field extensions, so is $K \subset F$.

Remark. The converse is also true.

$$\begin{array}{c} \alpha \in F \\ | \\ E \\ | \\ K \end{array}$$

Proof. Let $\alpha \in F$ be an element. Then $f(\alpha) = 0$ for some non-zero polynomial

$$f(t) = e_0 + e_1 t + e_2 t^2 + \ldots + e_n t^n$$

where $e_i \in E$ for all i. Consider a tower

$$\begin{tikzcd}
F \arrow[d, dash] \arrow[dr, dash] & \\
E \arrow[r, dash] \arrow[d, dash] & K(e_0, e_1, \ldots, e_n) \arrow[dl, dash] \\
K &
\end{tikzcd}$$

Note that α is algebraic over $K(e_0, e_1, \ldots, e_n)$ so that $K(e_0, e_1, \ldots, e_n, \alpha) \,/\, K(e_1, \ldots, e_n)$ is finite. Since $E \,/\, K$ is algebraic, it follows from proposition (the proposition tha tif $F \,/\, K$ is finite, $F = K(\alpha_1, \ldots, \alpha_r)$ for algebraic elements α_i) that $K(e_0, \ldots, e_n) \,/\, K$ is finite, hence algebraic. In particular, α is algebraic over K. $\square$

Proposition. (13.33) Let $E \subset F$ be the subset consisting of algebraic elements over K. Then E is a subfield of F.

Proof. Let α and β be algebraic elements over K. We show that $\alpha \pm \beta$, $\alpha\beta$, and α/β if $\beta \neq 0$ are all algebraic over K. Do this yourself using the above! $\square$

13.5 Applications: Geometric Impossibilies

Remark. Drawing shapes only using a straightedge and a compass. We know that two possible shapes are an equalateral triangle and a line perpendicular to the line passing through the middle point. What about the following?

(i) The side of a square with the same area as a circle

(ii) The side of a cube whose volume is 2

(iii) A line trisecting the angle formed by two lines

Definition. (13.36) A real number is constructable if it is a coordinate of a point one can construct by a straightedge and a compass:

(i) Draw a line through two points you have already constructed

(ii) Draw a circle with the center at a point you have constructed through another point you have constructed

(iii) Mark a point of interesection of two lines, a line and a circle, or two circles you have already constructed

Lecture 7 (September 23rd)

Remark. Last time we talked about constructible numbers. This class, we address some old questions on constructibility. Recall that we only allowed a straightedge (a ruler without marks) and compasses.

Definition. (13.36) A constructible number is obtained from the following operations:

(i) Making a straightline between two points

(ii) Drawing a circle around a point

(iii) Making interactions between lines and circles

Example. $\dfrac{1}{2}$ and $\dfrac{\sqrt{3}}{2}$ are constructible. In fact, all rational numbers are constructible. How about

(i) Squaring a circle, or $\sqrt{\pi}$

(ii) Doubling the cuble, or π

(iii) Trisecting the angle $60°$, or $20°$

Remark. If α and β are constructible real numbers, then so are $\alpha \pm \beta$, $\alpha\beta$, and α/β (if $\beta \neq 0$). Consequently, the set of constructible numbers is a subfield of $\boldsymbol{R}$ containing $\boldsymbol{Q}$.

Remark. If α is constructible, then so is its square root $\sqrt{\alpha}$.

Theorem. (13.38) Let α be a constructible number. Then $\alpha \in F$, where F is a field extension of $\boldsymbol{Q}$ such that $[F : Q]$ is a finite power of 2. In particular, α is an algebraic number and $[\boldsymbol{Q}(\alpha) : \boldsymbol{Q}]$ is a finite power of 2.

Proof. It suffices to show the following – in any stage of a construction, if all the constructed numbers lie in a field E, then applying one of the basic operations the same data lies in a field $E(u)$, where u is algebraic of degree at most over E.

Note that at any stage of a construction, we consider:

(i) A coordinate of points that have been constructed.

(ii) A coefficient in the equations of the lines and circles that have been constructed.

It is enough to intersect a circle and a line to get new constructible points. For this, we need to solve

$$\begin{cases} x^2 + y^2 + ax + by + c = 0 \\ y = mx + r \end{cases}$$

where $a, b, c \in E$ and $m, r \in E$. Notice that

$$x = \frac{-(mr + bm + a) \pm \sqrt{(2mr + bm + a)^2 - 4(m^2 + 1)(r^2 + br + c)}}{2(m^2 + 1)}$$

which is in $E(u)$ and $u = \Delta$. □

Corollary. (13.40) The three inquries from above are impossible. That is, none of the numbers $\sqrt{\pi}$, $2^{1/3}$, and $\cos 20°$ are constructible.

Proof. (I) If $\sqrt{\pi}$ is constructible, then so is π. However, as we expect all constructible numbers to be algebraic, and π is strictly not over $\boldsymbol{Q}$, the claim is false.

(II) If $2^{1/3} = \alpha$, the irreducible polynomial is $t^3 - 2 \in \boldsymbol{Q}[t]$. However,

$$\begin{aligned} [\boldsymbol{Q}(\alpha) : \boldsymbol{Q}] &= \deg(\operatorname{irr}(\alpha, \boldsymbol{Q})) \\ &= \deg(t^3 - 2) \\ &= 3 \end{aligned}$$

which is not a finite power of 2.

(III) For $\cos 20° = \alpha$, note that α is a zero of $8t^3 - 6t - 1 \in \boldsymbol{Q}[t]$. Again, the root test gureentees that the polynomial is irreducible over $\boldsymbol{Q}$. Same as the above, the degree of the polynomial is not a finite power of 2, and the number is not construtible. □

Lecture 8 (September 25th)

Remark. Last time, we've seen how $[\boldsymbol{Q}(\alpha) : \boldsymbol{Q}]$ is a finite power of 2 for a constructible α. Today, we'll learn about splitting fields.

Theorem. (13.38) Let α be a constructible number. Then,

(i) $\alpha \in F$, where $[F : \boldsymbol{Q}]$ is a finite power of 2

(ii) α is algebraic over $\boldsymbol{Q}$

(iii) $[\boldsymbol{Q}(\alpha), \boldsymbol{Q}]$ is again a finite power of 2

Theorem. To prove this theorem, it is enough to show that: $C = \{\text{numbers that have been constructed}\} \subset E$, a field, acted on by the basic operators gives $C \cup \{\text{newly constructed numbers}\} \subset E(u)$ where $[E(u) : E] \leq 2$. For example,

$$\{y = mx + r\} \cap \{x^2 + y^2 + ax + by + c = 0\}$$

solves to give

$$x' = \frac{-(2mr + bm + a \pm \sqrt{(2mr + bm + a)^2 - 4(m^2+1)(r^2 + br + c)})}{2(m^2+1)}$$

By setting $u = \sqrt{(2mr + bm + a)^2 - 4(m^2+1)(r^2 + br + c)}$, we find that both x' and $y' = mx' + r$ lies in $E(u)$. It is trivial from here that the field extension has a dimension lower than or equal to 2.

Chapter 14. Normal and Separable Extensions, and Splitting Fields

Theorem. (Kronecker) Let K be a field, and let $p(t) \in K[t]$ be irreducible. Then there exists an extension field that contains a zero α of $p(t)$.

Proof. Set

$$E = K[t]/(p(t))$$

□

Example. Let $K = \boldsymbol{R}$ and $p(t) = t^2 + 1 \in \boldsymbol{R}[t]$. The field of complex numbers can be defined as

$$\boldsymbol{C} = \boldsymbol{R}[t]/(t^2 + 1)$$

and would contain the root $\alpha = \bar{t}$. Note that

(i)
$$\bar{t}^2 + \bar{1} = \overline{t^2 + 1} = \bar{0} \in \boldsymbol{R}[t]/(t^2 + 1)$$

That is, $\bar{t}$ is a zero of

$$x^2 + 1 \in \Big(\boldsymbol{R}[t]/(t^2 + 1)\Big)[x]$$

(ii) Note that $\boldsymbol{R} \to \boldsymbol{R}[t] \to \boldsymbol{R}[t]/(t^2 + 1)$ is injective

(iii) $\boldsymbol{R}[t]/(t^2 + 1)$ is a field since $\boldsymbol{R}[t]$ is a PID and $t^2 + 1$ is irreducible in $\boldsymbol{R}$.

Lecture 9 (September 30th)

Definition. (14.2) Let $f(t) \in K[t]$. A splitting field for $f(t)$ is a field extension F/K such that $f(t)$ factors into linear terms

$$c \cdot \prod_{i=1}^{n} (t - \alpha_i)$$

with $F = K(\alpha_1, \alpha_2, \ldots, \alpha_n)$.

Definition. (13.19) A field F is algebraically closed if every nonconstant polynomial $f(t) \in F[t]$ has a zero in F.

Remark. If $K = \boldsymbol{Q}$, the existence of a the splitting field of $f(t)$ is immediate: if $f(t) = c \cdot \prod_{i=1}^{n}(t - \alpha_i)$ in $\boldsymbol{C}[t]$, then

$$F = K(\alpha_1, \ldots, \alpha_n) \subset \boldsymbol{C}$$

Theorem. (14.6) There exists a splitting field of $f(t) \in K[t]$.

Proof. Let simply lay out the idea. For any $f(t) \in K[t]$, $f(t) = p_1(t) f_1(t)$ where $p_1(t)$ is irreducible. By Kronecker's theorem, we can set $K_1 = K(\alpha_1)/K$, where $p_1(\alpha_1) = f(\alpha_1) = 0$. Then,

$$p_1(t) = p_2(t) f_3(t)$$

and we can set $K_2 = K_1(\alpha_2)$ where $p_2(\alpha_2) = 0$. Thus,

$$f(t) = (t - \alpha_1)(t - \alpha_2) g(t) \in K_2[t]$$

and so on. $\square$

Theorem. (14.3) Let K be a field. Then there exists an algebraic extension $\bar{K}/K$ such that $\bar{K}$ is algebraically closed.

$$\begin{tikzcd}
\bar{K} \arrow[dd, dash] \arrow[dr, dash] & \\
& L \arrow[dl, dash] \\
K &
\end{tikzcd}$$

Remark. Such an extension is unique: if F and L are two such extensions of K, then there exists an isomorphism $\sigma : F \to L$ such that $\sigma|_K = \mathrm{id}_K$ (that is, $\sigma(a) = a$ for all $a \in K$).

$$
\begin{tikzcd}
F \arrow[r, "\sigma"] & L \\
K \arrow[u] \arrow[r, "\mathbf{1}"] \arrow[ur, phantom, "\circlearrowright"] & K \arrow[u]
\end{tikzcd}
$$

Definition. (14.4) We call $\bar{K}$ the algebraic closure of K.

Lecture 10 (October 2nd)

Remark. Last class we talked about algebraically closed fields. This class we will be talking about isomorphism extensions theorem.

Definition. (13.19) A field K is algebraically closed if for every algebraic extension F of K, $F = K$, or equivalently, every nonconstant polynomial $f(t) \in K[t]$ splits into linear factors.

Definition. (14.4) $\bar{K}$ is an algebraic closure of K if

(i) $\bar{K}/K$ is algebraic

(ii) $\bar{K}$ is algebraically closed

Theorem. (Isomorphic extension) Let $\sigma : K_1 \to K_2$ be an isomorphism of fields. Let F/K_1 be an algebraic extension. Then there exists a field embedding $\bar{\sigma} : F \to \bar{K}_2$ extendeding σ (that is, $\bar{\sigma}|_{K_1} = \sigma$).

$$
\begin{tikzcd}
F \arrow[r, "\bar{\sigma}"] & \bar{K}_2 \\
K_1 \arrow[u] \arrow[r, "\sigma"] \arrow[ur, phantom, "\circlearrowright"] & K_2 \arrow[u]
\end{tikzcd}
$$

Proposition. (14.1) Let F/K be a field extension and let $p(x) \in K[x]$ be irreducible. Then there is a 1-1 correspondence between the sets of zeros of $p(t)$ in F and the set of K-embeddings $K[t]/(p(t)) \to F$ which extends the inclusion $K \hookrightarrow F$.

Proof. Suppose that $\alpha \in F$ is a zero of $p(t)$. Consider the evaluation homomorphism $\mathrm{ev}_\alpha : K[t] \to F$. Then, $\ker(\mathrm{ev}_\alpha) = (p(t))$. Conversely, if $\sigma : K[t]/(p(t)) \to F$ is a K-embedding, then $\sigma(\bar{t}) \in F$ is a zero of $p(t)$ in F.

$$
\begin{tikzcd}
K[t]/(p(t)) \arrow[r] & F \\
K \arrow[u] \arrow[r, "\mathbf{1}"] \arrow[ur, phantom, "\circlearrowright"] & K \arrow[u]
\end{tikzcd}
$$

$\square$

Remark. We now write $\sigma : R_1 \to R_2$ for a ring homomorphism. If $f(t) \in R_1[t]$ is a polynomial, we let $f_\sigma(t)$ denote the image of $f(t)$ under the induced homomorphism $R_1[t] \to R_2[t]$. Concretely, for $f(t) = \sum_{i=0}^n a_i t^i$ we have $f_\sigma(t) = \sum_{i=0}^n \sigma(a_i) t^i$.

Proposition. Let $\sigma : K_1 \to K_2$ be an isomorphism of fields. Let F_1/K_1 and F_2/K_2 be field extensions. Let $p(t)$ be an irreducible polynomial in $K_1[t]$. Suppose that $\alpha \in F_1$ is a zero of $p(t)$ and that $\beta \in F_2$ is a zero of $p_\sigma(t)$. Then there exists a unique isomorphism $\bar{\sigma} : K_1(\alpha) \to K_2(\beta)$ such that $\bar{\sigma}|_{K_1} = \sigma$ and that $\bar{\sigma}(\alpha) = \beta$.

$$
\begin{tikzcd}
F_1 & F_2 \\
K_1(\alpha) \arrow[u] \arrow[r, "\bar{\sigma}"] & K_2(\beta) \arrow[u] \\
K_1 \arrow[u] \arrow[r, "\sigma"] \arrow[ur, phantom, "\circlearrowright"] & K_2 \arrow[u]
\end{tikzcd}
$$

with irreducible

$$
\begin{cases} p(t) \in K_1[t] \\ p(\alpha) = 0 \end{cases} \qquad \& \qquad \begin{cases} p_\sigma(t) \in K_2[t] \\ p_\sigma(\beta) = 0 \end{cases}
$$

Proof. (I) Define $\bar{\mathbf{1}} : K(\alpha) \to K(\beta)$ by sending $f(\alpha)/g(\alpha) \in K(\alpha)$ to $f(\beta)/g(\beta) \in K(\beta)$.

(II) Define the function $\bar{\mathbf{1}}$ through

$$
\begin{tikzcd}
K[t]/(p(t)) & K[t]/(p(t)) \\
K(\alpha) \arrow[u] \arrow[r, "\bar{\mathbf{1}}"] & K(\beta) \arrow[u] \\
K \arrow[u] \arrow[r, "\mathbf{1}"] \arrow[ur, phantom, "\circlearrowright"] & K \arrow[u]
\end{tikzcd}
$$

Keeping this in mind, we write

$$\bar{\mathrm{ev}}_\beta \circ (\bar{\mathrm{ev}}_\alpha)^{-1}$$

$\square$